

Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

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ABSTRACT

Wheeled bipedal robots face a trade-off: the stiffness that buys precise standing shrinks the push-recovery envelope, while the integral action that removes equilibrium bias winds up during disturbances. The Anchored Conditionally-Gated Controller (ACC) answers both by *conditional activation*: a position integral confined to a 15 cm neighbourhood and enabled by an asymmetric envelope follower ($\tau_{\text{attack}} \approx 23$ ms, $\tau_{\text{release}} \approx 1.5$ s) that separates quiet stance from post-push ringdown. On a 10-DOF, 8.1 kg wheeled biped in MuJoCo (500 Hz physics, 100 Hz control) the architecture is dissected, not only demonstrated. A factorial ablation separates three coupled effects: gated damping produces the steadiness, the anchor integral the accuracy, and the gates are bit-identical to ACC in quiet stance yet decisive under disturbance. The push envelope is measurably chiral (-17.3 N), traced to the anti-symmetric hip-roll/hip-yaw channels. A closed-form model predicts the 27.0 mm standing offset to 0.92 % across a fortyfold gain sweep, as the price of a bounded integrator memory. ACC holds 0.294 ± 0.015 mm sagittal repeatability ($N=10$), 59-fold over this ablation's proportional-only rung, recovers omnidirectional pushes (108 N median, 82 N worst case), and stands 30 min at 0.0004 mm/min drift where six classical baselines and a full-state LQR all fall. A 2 ms-resolved delay screen prices deployment at a 6–8 ms latency cliff that one retuned gain moves outward.

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1. Introduction

Wheeled biped robots need millimeter-level standing precision, omnidirectional push recovery, and terrain adaptation, but no single classical controller integrates all three: LQR handles balance (Klemm et al., 2019), WBC posture (Klemm et al., 2020), and MPC terrain (Wang et al., 2025); locomotion, separately again, through learned policies (Hwangbo et al., 2019).

The difficulty is fundamental. Lowering k_p widens the capture basin at the cost of balance stiffness; omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a persistent, predominantly *sagittal* limit cycle regardless of k_p ($N=10$: 69.7 mm peak-to-peak sagittally against 13.4 mm laterally), on the one axis the wheels can drive; yet a global integral winds up during pushes.

We present the Anchored Conditionally-Gated Controller (ACC), which addresses both dilemmas through *conditional activation*. The core mechanism is a **proximity-gated anchor**: a position integral confined

to a 15 cm neighborhood, enabled by an **asymmetric envelope follower** ($\tau_{\text{attack}} \approx 23$ ms, $\tau_{\text{release}} \approx 1.5$ s) that distinguishes quiet stance from post-push ringdown. Coupled with a **gated damping boost**, ACC achieves 0.294 ± 0.015 mm sagittal repeatability under idealized conditions (clean sensors, 0 ms delay, $N=10$), at push performance equal to or better than a proportional-only baseline carrying no integral action. Absolute accuracy is a separate quantity: the robot parks 27.0 mm from the commanded home and holds that standoff to 0.009 mm.

A factorial ablation under a single protocol (Table 1) separates three effects the architecture couples: quiet-stance *steadiness* comes from the gated *damping* terms rather than from the anchor integral, which buys *accuracy* instead; the proximity gate and the asymmetric envelope are provably inert during quiet stance, so only a displacement experiment can measure them; and the pitch safety gate is load-bearing with no disturbance applied. The gates are enabling conditions that make

an otherwise inadmissible damping and integral configuration safe, not the proximate source of the precision. The object of study is a *structured classical prior*: a hand-designed, interpretable torque assembly. Supplying the nominal base action of a residual-learning stack (Johannink et al., 2019) is the downstream use it was designed for and is future work; no claim below depends on a residual policy. Classical controllers are therefore the comparison of record: six of them, four 4-state TWIP and two 6-state coupled variants, all fail, and a full-state LQR linearized from the plant itself survives two orders of magnitude longer and still fails, so platform complexity rather than model order or servo lag dominates the failure. A preliminary model-free PPO (Schulman et al., 2017) run is a *difficulty probe* and bounds nothing about what RL attains here.

The main contributions are:

- (1) A **proximity-gated anchor** with asymmetric envelope follower achieving sub-millimeter idle precision without sacrificing omnidirectional push survival.
- (2) A **two-channel conditionally-gated torque assembly** whose channel separation admits contact-loss recovery and per-leg terrain adaptation as independent extensions without retuning the anchor; both are summarised here and specified in full in the extended version.
- (3) A **factorial ablation** (additive L0→L3 + subtractive S0→S5) against six classical baselines, plus two control experiments that test the comparison itself: a full-state LQR derived from the plant rather than a reduced model, and a sweep of the coupled model’s one empirical constant.

2. Related Work

2.1. Wheeled Bipedal Platforms and Classical Control

Ascento (Klemm et al., 2019) demonstrated two-wheeled jumping with LQR balancing, later extended to WBC with kinematic loops (Klemm et al., 2020); SKATER (Wang et al., 2025) uses hierarchical MPC with terrain adaptation; comparable platforms pair a balance controller with separate yaw, height and roll loops. Whleaper (Zhu et al., 2024) is closest to our morphology (10 DOF, three hip DOFs per leg); it *switches* between LQR for balanced sliding and a separately trained PPO policy for walking by

locomotion mode rather than composing them on one control output. Composing a classical prior with a learned residual on one output is the residual-RL construction (Johannink et al., 2019), applied recently to wheeled-biped balance (Zhang et al., 2025); ACC supplies that prior. A recent review (X. Liu et al., 2024) notes the absence of a unified classical solution. Coupled pitch–roll LQR, standard on rigid two-wheeled platforms without articulated legs, reproduces on our morphology as a 6-state design and falls; so does the 4-state Direct Torque variant that removes the servo path, so actuation lag alone cannot account for the outcome (§5.4). Cascade control (Åström & Hägglund, 1995) and whole-body control via hierarchical inverse dynamics or task-priority QPs (Herzog et al., 2016) use fixed-gain nested or prioritized loops without conditional gates; gain scheduling (Rugh & Shamma, 2000) varies parameters continuously with one variable rather than responding to distinct physical regimes, the reason the task-priority QP benchmarked in §5.5 does not resolve the precision–recovery trade-off. Split-range, selector and override architectures (Åström & Hägglund, 2006) blend continuously in process control; ACC’s contribution is the synthesis of gating surfaces (spatial proximity, velocity envelope, pitch safety interlock) derived from wheeled-biped failure modes rather than generic process-variable limits.

2.2. Disturbance Rejection, Anti-Windup, and ACC’s Distinction

DOB (Ohnishi et al., 1996) and ADRC (Han, 2009) estimate disturbances temporally but cannot distinguish a push from post-push ringdown: both occupy the same frequency band. Classical anti-windup (back-calculation, conditional integration, the unified observer-based framework (Åström & Rundqwist, 1989)) triggers on *actuator saturation*; ACC’s anchor instead uses **spatial, physics-conditioned gating**, confining integration to a 15 cm neighborhood and separating quiet stance from ringdown by a velocity envelope, replacing the estimation-speed-versus-noise trade-off with an attack-versus-release one. Push recovery via capture point (Englsberger et al., 2015) and flight-phase control (Roscia et al., 2023) is complementary, and terrain adaptation is usually exteroceptive or learned (Lee et al., 2020); ACC extends proprioceptive contact-point estimation to

wheeled bipeds. Hyon & Cheng (2006) made a humanoid passive to unmeasured external forces by gravity compensation with optimal contact-force distribution; ACC meets that requirement by switching regime on proprioceptive gating variables instead.

3. Robot Platform

3.1. Robot Morphology and Simulation

The robot (Fig. 1) is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). Mass: 8.1 kg; thigh: 0.26 m; shin: 0.28 m; wheel radius: 0.06 m; hip width: 0.23 m. Torque limits: 30 Nm (HR, HY, WH), 150 Nm (HP, KN). MuJoCo (Todorov et al., 2012) runs at 500 Hz ($\Delta t_{\text{sim}} = 0.002$ s); ACC and every classical baseline at 100 Hz (5 substeps), with the baselines additionally measured at their 50 Hz design rate and the PPO reference at the 50 Hz rate it was trained at. All experiments use a 400 Nm/s torque rate limit and raw torque commands direct to actuators. Integrator: implicitfast; cone: pyramidal (elliptic for wheels, $\mu_s=1.2$, $\mu_k=0.8$ body); armature 0.008 kg·m² (wheels) and 0.02 kg·m² (leg joints); contact solref={0.002,1} (body), {0.005,0.7} (wheels). Softening solref[0] to 0.02 (tire-on-asphalt) raises planar idle RMS by ~60 % to ~1.2 mm, so every sub-millimeter idle figure is conditional on stiff ground contact.

Height convention

CoM-z, the whole-body centre-of-mass height above ground, is what ACC commands and regulates; the nominal standing pose is $h_{\text{CoM}}=0.404$ m. **Base-z**, the torso root-body height (qpos[2]), sits 13.2 cm above it in that pose, varying by less than 1 cm over the postures used here. All ACC results are in CoM-z; the PPO reference (§5.4) and the WBC baselines (§5.5) command base-z and are labelled accordingly, since substituting one convention for the other injects the full 13.2 cm offset as a reference error. ACC's posture map is calibrated at 0.354 m and 0.454 m and interpolated between them; outside that band joint targets are linearly extrapolated and bounded to [0.254, 0.554] m, the ± 10 cm bound the per-leg height split (§4.5) operates against.

4. ACC Formulation

4.1. Two-Channel Torque Assembly

ACC assembles a PD balance core, a proximity-gated anchor integral and posture regulation into two torque

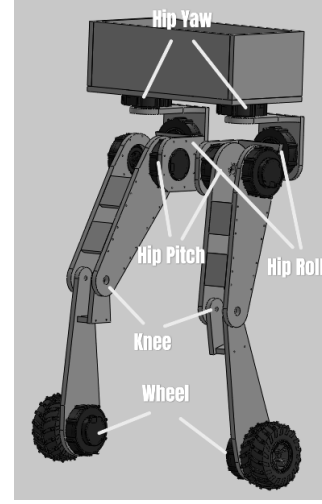


Figure 1. Robot joint layout with world frame (right-handed, Z up). Wheel axles lie along world X, so the rolling (and therefore sagittal) direction is world Y and world X is the lateral (track) axis. Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

channels, wheel torque ($\tau_w \in \mathbb{R}^2$) and leg-joint torque ($\tau_q \in \mathbb{R}^8$), with flight and terrain as independent extensions (Fig. 2):

$$\begin{aligned} \tau_w &= (1 - g_{\text{flight}}) [\tau_{\text{balance}} + g_{\text{anchor}} \tau_{\text{anchor}}] + g_{\text{flight}} \tau_{\text{flight}}, \\ \tau_q &= \tau_{\text{posture}}(h^{\text{cmd},L}, h^{\text{cmd},R}) + \tau_{\text{lat}} + \tau_{\text{yaw}}. \end{aligned} \quad (1)$$

Neither gate is a summand. At $g_{\text{flight}}=1$ the flight term *replaces* the wheel torque, since the balance core regulates against a ground that is no longer there, and the convex blend is what makes the replacement continuous. Per-leg ground adaptation likewise enters as a *reference* split: the posture regulator always takes two height commands, coincident on level ground, while g_{terrain} itself acts in the wheel channel.

Every gate g_* except the flight gate is a *smoothstep* function of a physical condition (proximity, quietness, attitude, terrain disparity) rather than a discrete switch:

$$ss(x; a, b) = 3t^2 - 2t^3, \quad t = \text{clip}\left(\frac{x-a}{b-a}, 0, 1\right). \quad (2)$$

Equation (2) is the standard Hermite blend, C^1 in x with zero derivative at both knees. Continuity matters concretely: a hard threshold injects a torque step equal to the full gated component, which at 100 Hz under a 400 Nm/s rate limit reaches the plant as an unmodelled impulse. The one exception is g_{flight} , and only on the way in: contact loss is a binary physical event, so engagement is a latched load-based hysteresis with asymmetric timers. Its handback is not discrete: after release the gate ramps linearly $1 \rightarrow 0$ over 150 ms, since

an instant handback at large pitch returns full wheel authority (~ 19 Nm) to a marginally loaded wheel and re-launches the robot.

Wheel torques control sagittal balance; lateral and yaw regulation actuate in the leg-joint channel through the antisymmetric hip-roll and hip-yaw pairs (Eq. (7)), the wheels carrying no roll or yaw differential. On level ground the channels are then coupled only through the plant, the terrain gate being identically zero. Across a step it opens, softening k_{pos} , raising the position-torque cap, adding a pitch-rate feedforward and symmetrising the per-wheel damping, but that coupling vanishes continuously with the height split, so flight recovery and terrain adaptation are still added without retuning the anchor. Unlike the industrial override architectures of §2.2, which select *among parallel controllers* on process-variable limits, ACC gates *components within a single torque assembly* on physical quantities of three kinds (spatial, temporal and safety-interlock), each chosen because a specific measured failure mode of this platform is expressible in it.

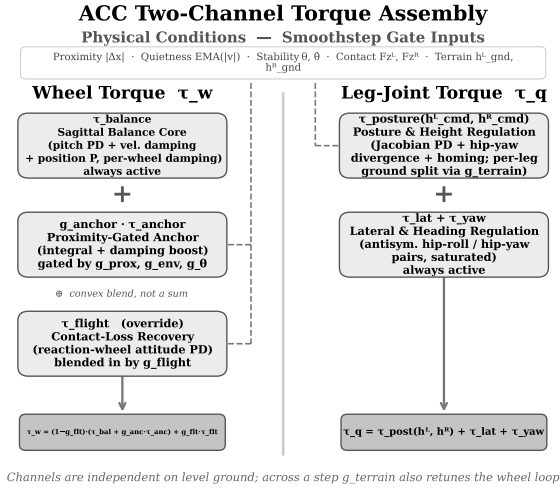


Figure 2. ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed). The latched flight gate *overrides* the wheel channel rather than adding to it, and per-leg ground adaptation enters as a split pair of height references, not an added torque.

4.2. Sagittal Balance Core (τ_{balance})

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\tau_{\text{balance}} = \begin{bmatrix} \tau_{\text{com}} + \tau_w^L \\ \tau_{\text{com}} + \tau_w^R \end{bmatrix}, \quad \begin{aligned} \tau_{\text{com}} &= \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}}, \\ \tau_w^i &= -k_w(\omega^i - \omega^{\text{cmd}}), \end{aligned} \quad (3)$$

with τ_{com} identical on both wheels and the per-wheel damping τ_w^i ($i \in \{L, R\}$) the only term that differs between them,

while lateral and yaw regulation, which read torso state alongside the balance core, actuate antisymmetrically on the hip-roll pair (HR^L, HR^R) and the hip-yaw pair (HY^L, HY^R) of the leg-joint channel:

$$\tau_{\text{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \quad (4)$$

$$\tau_{\text{damping}} = -(s_v k_{\text{vel}} + k_{v2}) \cdot v_{\text{sag}}, \quad (5)$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \text{clip}(\Delta x, \pm 0.10 \text{ m}), \quad (6)$$

$$\tau_{\text{lat}} = [1, -1]^T (k_{p,\text{roll}} \phi + k_{d,\text{roll}} \dot{\phi}), \quad (7)$$

$$\tau_{\text{yaw}} = [-1, 1]^T (k_{p,\text{yaw}} e_\psi - k_{d,\text{yaw}} \omega_z), \quad (8)$$

the last two saturated at their own moment limits, with s_v the scheduled damping scale, k_{v2} a fixed unscheduled damping offset, θ torso pitch, v_{sag} the sagittal body velocity, Δx the sagittal position error from the latched home position, ϕ the roll angle, e_ψ the yaw error and ω_z the yaw rate. The lateral axis is thus closed by the legs, and the fixed sign pattern of these two pairs is the mechanism behind the chirality of the push envelope. The P position term (k_{pos} , clipped to ± 0.10 m, saturating at ± 4 Nm) keeps the robot within $\pm 4-6$ cm of home but leaves a 1.3 Nm equilibrium bias sustaining a perpetual limit cycle, which motivates the anchor integral. Saturation-triggered anti-windup does not address it, the critical failure being *state-dependent*: during a push, position error grows to 10–30 cm while the wheels remain within their 20 rad/s range, so a saturation trigger never fires and ACC uses spatial confinement instead. Pitch stiffness is fixed at $k_p=50$; a scheduled variant softening it to 35 during pushes was tested and rejected (S0 vs. S1, Table 2).

4.3. Proximity-Gated Anchor ($g_{\text{anchor}} \tau_{\text{anchor}}$)

The anchor mechanism is the core contribution. It adds three terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\begin{aligned} \tau_{\text{anchor}} &= K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x) dt \\ &\quad + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}) \\ &\quad + (k_{d,h} g_h + k_{d,a} g_{\text{prox}}) g_{\dot{\theta}} \cdot (-\dot{\theta}), \end{aligned} \quad (9)$$

where the gates are continuous smoothstep functions equal to 1 when the robot is “at home and quiet” and transitioning smoothly to 0 otherwise:

$$g_{\text{prox}}(\Delta x) = 1 - \text{ss}(|\Delta x|; 0.05 \text{ m}, 0.15 \text{ m}), \quad (10)$$

$$g_{\text{env}}(v_{\text{sag}}) = 1 - \text{ss}(\text{EMA}(|v_{\text{sag}}|); 0.25 \text{ m/s}, 0.50 \text{ m/s}), \quad (11)$$

$$g_{\theta}(\theta, \dot{\theta}) = [1 - \text{ss}(|\theta|; 2^\circ, 12^\circ)] \times [1 - \text{ss}(|\dot{\theta}|; 2^\circ/\text{s}, 15^\circ/\text{s})], \quad (12)$$

$$g_h(\Delta h) = 1 - \text{ss}(|\Delta h|; 0.05 \text{ mm}, 0.30 \text{ mm}), \quad (13)$$

$$g_{\dot{\theta}}(\dot{\theta}) = \text{ss}(|\dot{\theta}|; 2^\circ/\text{s}, 15^\circ/\text{s}), \quad (14)$$

$$g_{\text{boost}} = g_{\text{prox}} \cdot g_{\text{env}} \cdot g_{\theta}. \quad (15)$$

Each gate is dimensionless, monotone and C^1 , and three physically distinct conditions must hold simultaneously for the anchor to integrate: spatially near home (g_{prox}), temporally quiet (g_{env}) and attitudinally settled (g_{θ}). The pitch gate closes when either pitch or pitch rate leaves its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope.

Asymmetric envelope follower. Reliable quiet-stance detection rests on an asymmetric exponentially-weighted moving average (EMA) of $|v_{\text{sag}}|$:

$$\text{EMA}_t = \begin{cases} \alpha_a |v_t| + (1 - \alpha_a) \text{EMA}_{t-1}, & |v_t| > \text{EMA}_{t-1}, \\ \alpha_r |v_t| + (1 - \alpha_r) \text{EMA}_{t-1}, & \text{else,} \end{cases} \quad (16)$$

with time constants $\tau_{\text{attack}} \approx 23 \text{ ms}$ and $\tau_{\text{release}} \approx 1.5 \text{ s}$ as the primary specification, the discrete coefficients following as $\alpha_a = 1 - e^{-\Delta t / \tau_a} \approx 0.35$ and $\alpha_r \approx 0.007$ at 100 Hz. The asymmetry is essential: a symmetric EMA is *bistable*: post-push oscillation holds it in a middle band where the gate is half-open and the boost cannot collapse the cycle, as the symmetric-EMA ablation of §5 confirms, whereas the fast-attack/slow-release envelope rises to 1 only when oscillation truly decays and drops to 0 within $\sim 25 \text{ ms}$ of a disturbance. The mechanism is a peak-hold envelope detector; ACC's contribution is coupling it to a spatial proximity gate for integral conditioning. Since $\text{EMA}_0 = 0$ biases $g_{\text{env}} \approx 1$ for $\sim 7.5 \text{ s}$ at startup, all measurements discard a 3–5 s settle interval.

4.4. Posture and Yaw Stability (τ_{posture})

Posture regulation uses Jacobian-based PD gains scheduled across 23 heights:

$$\tau_{\text{posture}} = \tau_{\text{PD}}^{\text{jac}}(q, \dot{q}, h^{\text{cmd}}) + \tau_{\text{div}}(q_{\text{hy}}^L, q_{\text{hy}}^R) + g_{\text{settled}} \tau_{\text{homing}}(q - q_{\text{ref}}), \quad (17)$$

where τ_{div} damps hip-yaw divergence and τ_{homing} restores nominal posture when upright and settled. Yaw

return uses wheel-differential torque.

4.5. Extensions: Contact Loss and Per-Leg Terrain

Two extensions attach to the same assembly without re-tuning the anchor, and both vanish identically in nominal standing. *Contact-loss recovery* (g_{flight}): when both wheels report $F_z < 0.5mg$ for two consecutive steps, ACC substitutes a flight attitude controller that uses the wheels as reaction flywheels, $\tau_{\text{flight}} = \text{clip}(2.0\theta + 0.5\dot{\theta} - 0.02\omega_{\text{wheel}}, \pm 2 \text{ Nm})$, with a five-step loaded-release hysteresis so balance torque cannot re-engage mid-bounce. *Per-leg ground adaptation* (g_{terrain}): a wheel resting at a different ground height would otherwise force the torso to roll by the step angle, so ACC maintains an independent ground estimate per leg from the wheel contact-point height (exact at any body tilt for a rigid wheel), updated only while that wheel carries at least $0.2mg$, and absorbs the disparity by a *kinematic* leg-length split plus a closed-loop levelling integrator trimming the residual by the measured roll. Both are specified in full, with their ablation matrices, in the extended version.

4.6. Complete Numerical Parameter Set

The core gains are $k_p = 50 \text{ Nm/rad}$, $k_d = 10 \text{ Nm/(rad/s)}$, $k_{\text{pos}} = 40 \text{ Nm/m}$ (clipped at $\pm 4 \text{ Nm}$), $k_{\text{vel}} = 15 \text{ Nm/(m/s)}$ at damping scale $s_v = 1.5$ plus a fixed $k_{v2} = 5 \text{ Nm/(m/s)}$, $k_w = 0.5 \text{ Nm/(rad/s)}$, $k_{\text{roll}} = 55 \text{ Nm/rad}$, $k_{\text{yaw}} = 1.0 \text{ Nm/rad}$; the anchor uses $K_i = 4.0 \text{ Nm/(m/s)}$ capped at 2 Nm with a 5×10^{-3} per-step leak, boost scale 5.0, gated pitch-rate gains $k_{d,h} = 3.0$ and $k_{d,a} = 7.0 \text{ Nm/(rad/s)}$ over a g_h band of 0.05–0.30 mm, g_{prox} band 0.05–0.15 m, g_{env} band 0.25–0.50 m/s, g_{stab} bands $2-12^\circ$ and $2-15^\circ \text{ s}^{-1}$; flight uses P/D = 2.0/0.5 with 0.02 Nm/(rad/s) wheel damping and 2/5-step engage/release at $F_z = 0.5mg$ (0.2mg for terrain). The full set (50+ gains) is in the supplementary material.

5. Experimental Validation

ACC and all six classical baselines are evaluated at 500 Hz physics / 100 Hz control under a 400 Nm/s per-joint torque rate limit, so the headline comparison is rate-matched by construction; the PPO reference is reported at the 50 Hz rate its policy was trained at. Each classical baseline was additionally measured at its own 50 Hz design rate: the fall rate is 100 % in all twelve cells and five of the six survive *longer*

at 50 Hz, so reporting them at 100 Hz is the rate-matched choice and the harsher of the two for them. Seeds are deterministic: 20260727+trial_id for idle standing, 20260728×100+rep for the push ablation, and {0, 42, 123} for the classical baselines.

5.1. Anchored Standing

5.1.1. Metrics and axis convention

“Standing precision” names two quantities that are not interchangeable. With $\mathbf{p}(t) \in \mathbb{R}^2$ the horizontal base position over a window of length T , $\mathbf{p}_0$ the commanded home and $\bar{\mathbf{p}}$ the trial mean,

$$A = \left(\frac{1}{T} \int_0^T \|\mathbf{p}(t) - \mathbf{p}_0\|^2 dt \right)^{1/2}, \quad (18)$$

$$R = \left(\frac{1}{T} \int_0^T \|\mathbf{p}(t) - \bar{\mathbf{p}}\|^2 dt \right)^{1/2} \quad (19)$$

define the *accuracy* A , which retains any DC parking offset, and the *repeatability* R , which removes it; the two differ by more than an order of magnitude here, so both are reported for every configuration. Each is resolved along the sagittal axis, the lateral axis and the full plane, the two horizontal axes being actuated by different means: the **sagittal** axis is world y , the rolling direction and the only one on which wheel torque produces ground force, while the **lateral** axis is world x , held passively by the wheel contacts and the roll restoring moment of the track width. A dynamic check confirms the assignment (scripts/verify_body_axes.py): a 90 N impulse along y is absorbed at a peak wheel speed of 52 rad/s leaving 4.8 mm of residual offset, whereas the same impulse along x leaves a permanent 23.2 mm offset. The anchor integral and the gated damping boost both live in the wheel-torque channel, so both act sagittally only.

5.1.2. Protocol

Every row of Table 1 is measured under one protocol: 25 s from the nominal standing pose with randomised initial conditions ($\sigma=0.005$ rad per joint, clipped to limits; $\sigma=1$ mm on root height), with A , R taken relative to each trial’s own $t=0$ pose after a settle covering the $\text{EMA}_0=0$ transient to within 1 %. A trial is a fall if $|\theta|>46^\circ$ or root height <0.15 m; confidence intervals are Student- t . This measures the ability to *stay* put; coming back is measured in §5.2.

5.1.3. Result

ACC holds **sub-millimetre sagittal repeatability**, $R_{\text{sag}} = 0.294 \pm 0.015$ mm (95% CI [0.283, 0.305] mm) across ten randomised trials with no falls. This is the

figure of merit for the platform, sagittal being the actuated inverted-pendulum axis. Precision is assessed from the relative spread *between* configurations under an identical protocol rather than against a simulator noise floor: removing a single mechanism (S2) moves R_{sag} sixteenfold with non-overlapping SDs. Against the P-only baseline L0, which sustains a 69.7 mm peak-to-peak sagittal limit cycle, the like-for-like improvements are **59.4×** sagittal, **22.6×** planar and **5.0×** lateral, largest on the hard axis and smallest on the easy one, the correct signature for a wheel-torque controller, since every term ACC adds beyond L0 produces ground force along the rolling direction and none can produce lateral force at all.

Accuracy and repeatability are distinct on this axis. ACC parks at $\bar{s} = -27.0$ mm from commanded home and holds that standoff to 0.009 mm SD across trials, so planar accuracy is $A_{xy} = 26.994$ mm, a 1.8× improvement over L0 against the 59× gain in repeatability. Its magnitude is fully accounted for: instrumenting the sagittal torque balance predicts the offset in closed form from a *leaky* anchor integral whose ~ 2 s forgetting horizon fixes its DC gain at the finite $k_{I,\text{dc}}=7.96$ Nm/m, so a constant load leaves a finite error by construction, and a 1.44° error in the feedforward pitch reference, giving 24.88 mm against 24.99 mm measured and holding across a 40× gain sweep and a five-point height sweep to within 0.92 %. The leak also buys the steadiness: a 10× smaller λ cuts the offset to -16.31 mm but degrades window jitter twelvefold, so Table 1 publishes a chosen point on that trade-off.

5.1.4. The measurement window is a protocol choice, not a stability horizon

Every row of Table 1 is measured over 20 s, and that standard cuts both ways. Re-running the ACC row with identical seeds, initial conditions, profile and metrics, and only the window lengthened, reproduces 0.294 ± 0.015 mm at 20 s and then *improves* on it, to 0.089 ± 0.005 mm at 300 s (10/10, drift $+0.0149$ mm/min) and 0.037 ± 0.002 mm at 1800 s (4/4, $+0.0004$ mm/min): the 20 s figure is the conservative one, that window still containing the tail of the settle. The settle terminates in a fixed point, not a slow mode: after ≈ 35 s the sagittal coordinate is constant to 2×10^{-6} mm per 30 s block for the remaining twenty-nine minutes at a -26.763 mm parking offset, the only motion left a 0.14 mm/hour lateral creep on the passively held axis. The full-state LQR of

Table 1. Idle standing precision across the full ablation ladder. All rows measured under the single protocol of §5.1; values are mean±SD over $N=10$ trials.

Configuration		R_{sag} (mm) sagittal rep.	R_{xy} (mm) planar rep.	R_{lat} (mm) lateral rep.	$\bar{s}$ (mm) sag. offset	Survival
Additive ladder (bottom-up)						
L0	P-only (PD, no position, no integral)	17.461 ± 0.258	17.845 ± 0.279	3.675 ± 0.246	−44.8	10/10
L1	+ position homing (±0.10 m)	20.397 ± 0.017	20.414 ± 0.019	0.834 ± 0.066	−30.4	10/10
L2	+ ungated integral K_i	25.026 ± 0.039	25.056 ± 0.043	1.212 ± 0.110	−27.5	10/10
L2.5	+ proximity gate only (no envelope, no boost)	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	10/10
Proposed controller						
L3/S1	ACC (full anchor, fixed $k_p=50$)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	10/10
Subtractive ablation (top-down from ACC)						
S2	− gated damping boost	4.772 ± 0.069	4.824 ± 0.067	0.703 ± 0.069	−26.8	10/10
S3	− asymmetric envelope (symmetric EMA)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	10/10
S4	− proximity gate (integral always on)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	10/10
S5	− pitch safety gate g_θ	—	—	—	—	0/10
Single-mechanism isolation (ACC with one term disabled)						
M1	$K_i=0$ (anchor integral removed, gates intact)	0.089 ± 0.035	0.733 ± 0.072	0.727 ± 0.071	−31.2	10/10
M2	damping terms reverted to their L2 values	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	10/10
M3	$K_i=0$ and boost = 0	3.973 ± 0.130	4.037 ± 0.128	0.710 ± 0.069	−31.1	10/10

100Hz control, 20 s window after a 5 s settle, clean sensors, 0ms delay. R = repeatability (Eq. (19)); $\bar{s}$ = mean signed sagittal parking offset; accuracy on an axis is that offset and the repeatability in quadrature, so the A columns are omitted (the identity holds to 0.003 mm on every surviving row). Sagittal is world y , lateral world x ; the anchor integral and the damping boost act sagittally only, so R_{sag} and $\bar{s}$ discriminate between configurations while R_{lat} measures a passively held axis. Produced by `scripts/collect_idle_ladder.py`.

Section 5.4.1, by contrast, accumulates 0.43–0.63 m of planar drift and falls, having no term that references position to a latched home; ACC has two, the clipped proportional τ_{pos} of Eq. (3) and the anchor integral on $-\Delta x$ of Eq. (9).¹

5.1.5. Which mechanism produces the precision

The lower blocks of Table 1 isolate the contributions; three of the four findings contradict the attribution the gate structure invites. (i) *Idle steadiness comes from the damping ladder; the integral buys accuracy instead.* M1 disables the anchor integral ($K_i=0$) with every gate in place, and its repeatability is slightly better than ACC's while its accuracy is worse by 4.2 mm; M2 keeps the integral and every gate but reverts the two velocity-damping coefficients to their L2 values, reproducing the L2/L2.5 limit cycle exactly. Steadiness is therefore attributable to the damping terms, in two stages: raising the velocity-damping scale takes R_{sag} to 3.97 mm (M3), the gated boost the rest of the way, a combined factor of 86. (ii) *The damping boost acts sagittally, as it must.* S2 degrades sagittal repeatability sixteenfold while leaving lateral repeatability statistically unchanged (overlapping CIs), the only outcome consistent with a boost inside the wheel-torque channel.

(iii) *The proximity gate and the asymmetric envelope*

are inert during quiet stance. Ablating either (S4, S3) is *bit-identical* to ACC on every idle metric, because the base never leaves the 5 cm inner band and the velocity envelope never rises into the attack branch, so both are evaluated only in Table 2. (iv) *The pitch safety gate is required even to stand.* S5 falls 10/10 during quiet stance with no push applied: removing g_θ lets the damping boost act on the pitch state at large excursions, and the resulting positive feedback diverges from the ±0.005 rad initial perturbation alone.

5.2. Omnidirectional Push Recovery

Push recovery is evaluated by polar sweep: 10–160 N impulses of 7 control steps applied at the torso in 8 world-frame directions ($\theta = \pm 90^\circ$ sagittal, $\theta = 0^\circ/180^\circ$ lateral), each threshold located by binary search to 5 N, survival being torso pitch $< 46^\circ$ for 17 s post-push; F_{min} is the maximum magnitude survived in *every* direction, F_{med} the median.

The envelope is anisotropic by construction. Per-bearing means span 82.1–144.1 N: the two lateral bearings average 98.0 N against 130.0 N for the two sagittal ones, the weakest being the oblique 45° (82.1 N, 24 % below F_{med}). This follows from the morphology, not the tuning: the wheels drive the contact patch back under the CoM sagittally but produce no lateral ground force, so raising the lateral bearing requires a centralised

¹Seeds 20260727+*i*, 5 s settle, clean sensors, 0ms delay, as in Table 1; produced by `scripts/idle_longhorizon.py`.

lateral task (§5.5), not a gain change.

The envelope is also chiral, and the chirality is in the controller. Superposed on the anisotropy is a left–right asymmetry: 135° survives 40.7 N more than -135° . The plant cannot produce it: mirroring the model about the sagittal plane reproduces its mass, inertia and geometry to 0.05 mm and 1×10^{-7} , and a harmonic fit localises the violation in the mirror-forbidden term ($b_2 = -17.3$ N against $b_1 = 0.3$ N). Under a mirrored state the commanded torque maps to its own mirror image on hip-pitch, knee and wheel and fails only on hip-roll and hip-yaw, both assembled antisymmetrically; mirroring those two channels flips b_2 to 15.6 N at unchanged magnitude, a sham mirror leaving it at -15.7 N. The correction is reported but not adopted, redistributing the envelope without enlarging it ($F_{\min}$ 82.5 $\rightarrow$ 79.7 N). Later protocols push along world $+x$, a lateral and below-median bearing (95.5 N), so they are conservative probes.

Table 2 reports the factorial ablation. Additively, only homing (L1) and the full anchor (L3) add significant margin; the global integral and the proximity gate add nothing to push *margin* on their own (the gate prevents windup, it does not enlarge the capture basin) and the full anchor is the only rung that settles. Subtractively the ablation is flat: at $N=30$ every single-mechanism removal except the pitch gate leaves $F_{\min}$ within 1.4 N of ACC ($|d| \leq 0.49$), so the capture basin is set by the anchor as a whole, not by any one of its four gates, including gain scheduling (S0) and the damping boost, which dominates idle precision yet leaves push margin indistinguishable from ACC’s (S2). Removing the pitch gate (S5) leaves the robot unable to survive even the search floor. Ringdown decays within 9 s under ACC, whereas P-only oscillates indefinitely on the 1.3 Nm equilibrium bias; symmetric EMA shows the bistable failure the asymmetry prevents, and the ungated integral winds up during the push and never settles.

5.3. Flight Recovery and Terrain Adaptation

At $N=10$ randomised-IC trials, ACC settles 10/10 from every drop and ledge tested (peak $|\theta|$ of $24.0 \pm 1.2^\circ$ on a 100 cm drop and $27.1 \pm 2.0^\circ$ on a 50 cm ledge) and traverses 10/10 at every curb height to 20 cm (0.15 m/s) at a straddle roll of $2.4 \pm 0.1^\circ$ to $3.9 \pm 0.1^\circ$. Replacing the per-leg split with a symmetric posture falls at *every* curb height.

Table 2. Factorial ablation of ACC components: push-recovery and ringdown metrics. Clean sensors, all rows $N=30$, mean $\pm$ std.

	Configuration	$F_{\min}$ (N)	F_{med} (N)	Ringdown
Additive (bottom-up)				
L0	P-only (PD, no pos., no I)	70.0 $\pm$ 1.3	122.8 $\pm$ 3.3	never
L1	+ Pos. homing (± 0.10 m)	75.0 $\pm$ 1.1	130.5 $\pm$ 1.9	never
L2	+ Integral K_i (no gate)	74.9 $\pm$ 0.9	131.5 $\pm$ 1.4	—
L2.5	+ Prox. gate only (no EMA, no boost)	74.9 $\pm$ 0.9	131.4 $\pm$ 2.0	—
L3	+ Full anchor (gate, EMA, boost)	82.1$\pm$2.7	108.4 $\pm$ 6.6	9.0 s
Subtractive (top-down from ACC)				
S0	ACC + scheduled k_p (rejected)	80.7 $\pm$ 2.8	110.2 $\pm$ 6.2	9.0 s
S1	ACC (canonical, fixed $k_p=50$)	82.1$\pm$2.7	108.4 $\pm$ 6.6	9.0 s
S2	–Damping boost	81.2 $\pm$ 1.0	115.9 $\pm$ 6.2	never
S3	–Asym. EMA (sym. EMA)	81.5 $\pm$ 2.6	106.1 $\pm$ 6.2	never
S4	–Prox. gate (global I)	80.7 $\pm$ 2.8	110.2 $\pm$ 6.2	windup
S5	–Pitch gate g_θ	<10	<10	never

$F_{\min}$ and F_{med} are the minimum and median survivable push over 8 bearings, each located by binary search to 5 N tolerance; idle metrics for the same configurations are in Table 1. The S5 entries are left-censored: in 231 of 240 trials the bisection found no survivable magnitude in $[10\text{N}, 160\text{N}]$ and the nine exceptions reach at most 46.3 N, so both are upper bounds. Welch’s unpaired two-sided t -test ($N=30$ per group, Bonferroni-adjusted $\alpha=0.0056$, 9 comparisons): L0 $\rightarrow$ L1 +5.1 N, $p=2 \times 10^{-23}$, $d=4.3$; L1 $\rightarrow$ L2 $p=0.69$; L2 $\rightarrow$ L2.5 $p=0.86$; L2.5 $\rightarrow$ L3 +7.2 N, $p=1 \times 10^{-15}$, $d=3.6$. No subtractive row differs significantly from ACC except S5 ($p=2 \times 10^{-44}$), and those four nulls are bounded rather than underpowered ($|d| \leq 0.49$, <1.4 N on $F_{\min}$). The 95% CI half-width is $\approx 0.37 \times \text{std}$, so the push rows resolve ≈ 1 N. Each trial starts from a freshly constructed controller context. Produced by `scripts/replicate_ablation_n30.py`.

5.4. Comparison with Classical Baselines and a Preliminary RL Reference

No balance controller has been published for this morphology, so no established classical design exists for this plant to compare against. What can be measured instead is *transfer*: how far the reduced-order controllers the wheeled-biped literature does publish carry when applied here. Read the rows below as a transfer result rather than as a ranking of ACC against optimal control; the row that tests the linear-design family on its own terms is the full-state controller of §5.4.1. Six classical baselines are evaluated under identical environment setups: four derived from the 4-state TWIP model (Grasser et al., 2002) (LQR with decoupled roll/yaw PD through a PID servo layer of ~ 0.25 s lag;

the same plus a pitch-error integral with conditional anti-windup; a direct-torque variant re-optimised for the zero-lag plant; and PI+AW, adding a conditional forward-position integral) and two 6-state coupled variants adding the roll states ($\phi, \dot{\phi}$), one through the servo and one commanding torque directly, all weights following Bryson's rule (Bryson & Ho, 1975). A model-free PPO run (5.4M steps, single seed, no domain randomisation or hyperparameter search) is reported alongside as a difficulty probe, not a tuned RL baseline (Limitation 4).

Table 3. Classical baselines and the preliminary PPO reference vs. ACC, clean sensors. All rows at 100 Hz control except the PPO reference, which is reported at the 50 Hz rate its policy was trained at.

Configuration	Surv. (s)	θ_{RMS} (°)	ϕ_{RMS} (°)	H_{RMS} (mm)	τ_{RMS} (Nm)
LQR, TWIP ($N=20$)	0.32	0.006	24.2	205.8	21.99
LQR + AW ($N=20$)	0.32	0.006	24.2	205.8	21.99
LQR, DT (re-opt.) ($N=20$)	0.60	0.033	27.5	119.2	8.76
6-St. LQR ($N=20$)	0.32	0.006	24.2	205.8	21.99
PI + AW, DT ($N=20$)	0.52	0.181	26.1	171.3	10.42
6-St. LQR, DT ($N=20$)	0.16	0.001	21.2	128.5	14.44
PPO ref. (50 Hz) ($N=20$)	3.20	4.82	3.15	54.2	18.40
ACC ($N=10$)	20.00	0.060	0.010	0.18	3.28

LQR / LQR+AW / 6-St. use the PID servo; the DT rows command torque directly. The three servo rows are identical because that servo's response dominates before the gain differential between them can act. Survival time is deterministic on all six classical rows (± 0.00 s at $N=20$, seeds {0, 42, 123}).

All six classical baselines fall within 0.60 s. Removing the PID servo path cuts torque $2.5\times$ and nearly doubles survival but *worsens* roll: the recovered authority is spent in the sagittal plane the model describes, not in the lateral one that ends the episode. The family is swept rather than tuned once: a 46-configuration retuning sweep leaves the outcome unchanged, as does an $80\times$ sweep of the coupled model's one empirical constant $b_{r,h}$ (hip-roll-to-roll-acceleration coupling), which falls at 0.320 s for every value though the re-derived roll gain varies $32\times$. The mechanism is saturation: the roll channel is commanded against the ± 0.7 rad hip-roll limit throughout every episode. The preliminary PPO reference stands longer at a single height but falls in 85 % of multi-height command transitions.

Every episode ends on roll. None of these controllers fails at the task LQR is credited with on Ascento (Klemm et al., 2019) or DIABLO (D. Liu et al., 2024): the servo variants hold pitch to 0.006° RMS and the full-state design of §5.4.1 holds 0.12° for the full 20 s. What terminates every episode is *roll*, on a

morphology carrying two lateral degrees of freedom per leg that those platforms lack, and the published designs do not ask a single LQR to close that axis either, DIABLO running separate yaw, split-angle, height and roll loops (D. Liu et al., 2024). The failure is therefore morphological: reduced-order LQR of the form the wheeled-biped literature uses does not transfer to a 10-DOF articulated-leg plant *as a single controller for all axes*. ACC's proportional-only rung L0 stands 10/10 (Table 1) where an optimal design does not because it keeps the dedicated roll, yaw and posture channels no row of Table 3 has: one linearisation for all axes against per-axis structure, not optimal against hand-tuned.

5.4.1. A full-state LQR designed on the plant itself

All six baselines are designed on a reduced model, confounding method with model. The control experiment that separates them derives a regulator from the plant directly: a damped Gauss–Newton solve for a true static equilibrium at the commanded height, central differences on the whole held-input control step of the complete 16-DOF, 10-actuator MuJoCo model for the transition Jacobians, and a discrete-time Riccati solve on the 30 states left after removing the two cyclic wheel angles, with Bryson's rule fixing $\mathbf{Q}$, $\mathbf{R}$ and a uniform control weight r swept over six decades. It is sound on every diagnostic that bears on it: residual unbalanced force 0.428 N against 79.46 N of body weight, $\mathbf{A}$ of full rank 30 and PBH-stabilisable, Riccati relative residual $\sim 10^{-14}$, closed-loop $\max |\lambda| = 0.9911$, and $\kappa(\mathbf{A}) \approx 10^{10}$ a 50 Hz artifact (2.6×10^7 at 100 Hz) from contracting contact modes that obstructs neither solve nor feedback. What the sweep shows is a slow divergence a short horizon hides: over 20 s the design looks successful ($r=1000$ at 0.65 m: 0 % falls, 11.0 mm height RMSE), while at 60 s every configuration falls in every episode at 22.2–57.0 s, ended by planar drift growing monotonically to 0.43–0.63 m, not by attitude. Nor does one weight cover the band: $r=100$ stands 20 s at 0.60 m and 0.69 m but falls at 0.65 m and $r=1000$ does the reverse, the fall rate runs 45–80 % under random height commands, and the maximum recoverable push is 10.0–20.2 N against ACC's $F_{\min}=82$ N. Full-state optimal linear feedback about a single equilibrium therefore buys two orders of magnitude of survival over the reduced-order designs and still does not close the problem. ACC, held to the same lengthened horizon, returns the opposite verdict: 10/10 at 300 s, 4/4

at 1800 s, residual drift 0.0004 mm/min against this design’s 0.43–0.63 m before falling (§5.1.4).

5.5. Whole-Body Control Baselines

Whether a tuned whole-body controller could match ACC was tested over 225 scenarios spanning fixed-height balance, height transitions and push recovery at 20–120 N under cloned initial conditions (481 000 control steps, no gain retuning), the task-priority quadratic-program WBC having been verified beforehand against MuJoCo’s `mj_jac`, with four selectable task weightings and ACC’s CoM- z height convention. As the *primary* balance controller it falls in **225 of 225** scenarios, where ACC falls in none: it holds the robot upright for 0.570 s and fails *laterally* (roll RMS 93.45° against ACC’s 3.71°). Neither compute nor task weighting is the constraint (the QP costs 0.169 ms mean, OSQP (Stellato et al., 2020)); the mechanism is that the posture task is a velocity damper rather than a position tracker, its reference rebuilt from the measured joint vector every step, so it collapses to $-k_{d,\text{posture}}\dot{\mathbf{q}}$ and can slow a postural drift but not reverse one. Blended instead as an assist on ACC, an ungated $\alpha=0.25$ never falls but degrades pitch RMS 0.52°→1.295° and planar drift 2.0×; gated, the admitted authority is $\bar{\alpha}=0.0061$. Roll is the exception, improving to 3.20° because the WBC closes it with a centralised orientation task where ACC uses decentralised hip-roll PD, a credible direction for ACC’s weakest axis, the 98.0 N mean lateral bearing of §5.2, but not through torque-level blending with a posture stack that cannot hold a posture.

5.6. Architectural Analysis

Robustness to sensor noise and actuator delay. A factorial sweep over four noise and three delay levels ($N=20$ per cell, 240 trials) bounds ACC on both axes at once (Table 4). One control period of delay is benign for stability and expensive for performance: at 10 ms there are zero falls on clean and low noise, but $F_{\max}$ nearly halves and idle RMS rises 5.6×. Three periods is a different regime, the platform falling 20/20 on *clean* sensing at 30 ms. Noise alone is survivable to the medium level; the *conjunction* is not: medium noise with 10 ms of delay produces the only sub-catastrophic falls in the grid. High noise is unrecoverable at every delay including zero, because the asymmetric envelope cannot distinguish noise from motion, the same mechanism behind the sharpest single sensitivity, idle repeatability degrading 71-fold under medium noise at zero delay as

the envelope gate releases and re-engages the integral. ACC is therefore specified against a filtered velocity estimate, a requirement on the state estimator rather than on the controller.

Table 4. Robustness to sensor noise and actuator delay, $N=20$ per cell. Idle RMS: lateral (world x) CoM std. over 20 s after a 3 s settle, *surviving trials only*. $F_{\max}$: binary-search lateral (world $+x$) push, ± 5 N tolerance, floor 10 N; delay is quantised to the control period here, and resolved more finely in the text.

Condition	Falls	Idle RMS (mm)		$F_{\max}$ (N)	
		Mean	$\pm$ Std	Mean	$\pm$ Std
Clean, 0 ms delay	0/20	0.43	0.06	95.7	0.8
Clean, 10 ms delay	0/20	2.43	0.21	50.5	7.9
Clean, 30 ms delay	20/20	— (all fell)		—	
Low noise, 0 ms delay	0/20	5.86	2.53	96.0	1.0
Low noise, 10 ms delay	0/20	10.66	3.70	51.3	8.9
Low noise, 30 ms delay	20/20	— (all fell)		—	
Med noise, 0 ms delay	0/20	30.82	12.13	93.5	5.2
Med noise, 10 ms delay	2/20	35.38	21.63	51.3	13.4
Med noise, 30 ms delay	20/20	— (all fell)		—	
High noise, all three delays	20/20	— (all fell)		—	

Noise ($1-\sigma$): low = 0.01 rad/s angular velocity, 0.01 m/s² gravity, 0.001 rad joint position; med = 5× low; high = 10× low. Delay is a transport lag on the actuator command; high noise fails identically at all three delays, so those rows are merged. The idle column averages survivors only, so the medium-noise 10 ms row is mildly optimistic and its falls column is primary. Produced by `scripts/collect_robustness_sweep.py`.

Resolving the delay axis. The grid quantises delay to the control period, so the knee was located separately with the transport delay applied per *physics* substep (2 ms at 500 Hz). It lies between 6 ms and 8 ms: $F_{\max}$ is flat to within 2.4 % out to 6 ms (93.2–95.5 N), then collapses 45 % to 51.0 N and declines monotonically to the 10 N search floor by 20 ms, a genuine loss of margin rather than a narrow phase-alignment notch, and the 6.6–8 ms end-to-end budget of §6 lands inside that bracket. Losing push margin and losing the platform are different thresholds: the standing threshold is 14–16 ms, station held through 14 ms at 10.84 mm idle lateral RMS. The knee is not an exhausted margin: at 8 ms each of four profile constants sits in a sharp *local dip* at its shipped value (51.0 N, nearly every neighbouring value recovering 78–93 N), and doubling the scheduled sagittal damping $s_p k_{\text{vel}}$ to 45 Nm/(m/s), denoted ACC-DH, moves the knee outward to 8–10 ms (51.0 → 93.2 N at 8 ms) for +1.4 % idle lateral RMS. It is an evaluation variant, not adopted: that constant also governs driving acceleration and settling.

Gate sensitivity and rate-limit invariance. Finite-difference gradients give a stark hierarchy: the release coefficient α_r dominates (sensitivity 220.4, 20 $\times$ the next and 130 $\times$ the pitch gates), the pitch thresholds being insensitive safety interlocks. The three gate failure modes map onto measured rows of Table 2: stuck-at-1 causes windup and a fall (S4), stuck-at-0 the P-only limit cycle (L0), stuck-half-open the parametric resonance of S5. The 400 Nm/s slew limit does not set the push ceiling: over an 8 $\times$ span lateral $F_{\max}$ moves only from 92.5 ± 1.1 N at 100 Nm/s to 94.4 ± 0.0 N at 200–800 Nm/s while the commanded slew saturates exactly at each bound, so the ceiling is architectural.

Compound disturbance. A commanded height transition and a push compete for the same leg-torque budget, so the standard lateral impulse is applied at the midpoint of a 1 s height ramp of size Δh about the 0.404 m nominal, with $\Delta h=0$ as the matched static control ($N=10$ per row). Inside the calibrated band a simultaneous height command is nearly free (+5 cm costs 5.1 % of push margin; –5 cm is *better* than static, +6.5 %), and the degradation outside it is one-sided: –10 cm costs nothing measurable ($p=0.99$) while +10 cm costs 68.7 %, since rising drives the knee toward extension, where the leg Jacobian is worst conditioned, while raising the CoM; re-parameterising the same posture family so achieved height matches the command closes 35 % of that deficit ($p=1.5 \times 10^{-4}$). Sequential disturbance is absorbed cleanly: a 90 N impulse followed 2 s later, before the first ringdown has decayed, by a 60 N counter-directed one is survived **10/10** at peak pitch $11.89 \pm 0.51^\circ$, which is what $\tau_{\text{release}} \approx 1.5$ s is designed to produce, the damping boost still partly engaged when the second impulse arrives.

5.7. Local Stability Certificate at the Standing Fixed Point

ACC's gates make it a switched system, of the kind partitioned by discrete switching surfaces and softened by hysteresis (Liberzon, 2003; Hespanha et al., 2003). Everything reported so far is behavioral evidence rather than a stability statement; this section supplies one.

At quiet stance the switching is inactive, by a measured margin. Over the last 20 s of a 60 s settle, 19 of the 21 gates the controller exports sit at exactly 0 or exactly 1 at every control step: bit-exact saturation, not approximate. The proximity gate's argument $|e_{\text{sag}}|$ peaks at 24.8 mm against its 50 mm knee (16.1–33.2 mm across the five

height variants), and the envelope gate's argument peaks at 1.5×10^{-7} m/s against its 0.25 m/s knee, a margin of 1.7×10^6 . The two gates that do sit inside their bands, anti-twist at ≈ 0.53 and centering at ≈ 0.986 , are ordinary smooth feedback, a smoothstep being C^∞ in the interior of its band. In a neighbourhood of the fixed point every branch selection is therefore frozen, the closed loop is a single smooth map, and Lyapunov's indirect method applies: chattering is avoided not because the gates are C^1 but because the operating point stands the above margins away from every switching surface.

The one switch that does fire is a contraction on both branches. The asymmetric envelope of Eq. (16) selects $\alpha \in \{0.35, 0.0067\}$ on the sign of $|v_{\text{sag}}| - \text{EMA}_{t-1}$, which in quiet stance flips at essentially every step, so the composition is only C^0 in time. For a fixed input level the tracking error $e_t = \text{EMA}_t - |v_{\text{sag}}|$ obeys $e_{t+1} = (1 - \alpha_t)e_t$, so $|e_{t+1}| \leq 0.9933 |e_t|$ under *either* branch: $V(e) = e^2$ is a common Lyapunov function for the switched pair, and the envelope is globally exponentially stable under arbitrary switching, worst-case rate set by the slow branch ($\tau=1.49$ s). Switching among individually stable subsystems therefore cannot destabilize the whole (Liberzon, 2003) and no dwell-time or hysteresis condition is required (Hespanha et al., 2003); in any case the discontinuity reaches the loop only through a gate that is flat ($\equiv 1$) across the entire quiet band.

The linearized closed loop. We central-difference the whole 100 Hz control step (control law plus five physics substeps) about the settled state, over an augmented vector of 57 coordinates: 30 plant (16 velocities and 14 configuration tangents, the two cyclic wheel angles removed as in §5.4.1), 3 of estimator memory and 24 of controller memory, with the adaptive-bias trim held at its converged value so that the one-step map is time-invariant rather than 300-periodic. At all five heights the spectrum has exactly two eigenvalues on the unit circle, none outside it, and a reduced spectral radius of $0.99787 - 0.99818$, a slowest stable mode of 4.7–5.5 s (Table 5). The classification is not a threshold judgement: the gap from the unit circle to the next mode is 1.8×10^{-3} , four orders above the spectrum's 10^{-7} sensitivity to the differencing step.

Both centre directions are identified and bounded by measurement. The first is lateral translation: a differential-drive base produces no lateral force to first order, a non-holonomic centre direction at $|\lambda|=1$ to

10^{-12} . The second is deliberate: the heading channel carries an explicit deadband, $\text{ss}(|e_{\text{yaw}}|; 0.02 \text{ rad}, 0.08 \text{ rad})$, measured at exactly 0 here, so absolute yaw enters no control law while the error stays inside $\pm 1.15^\circ$. Neither is left to the linearization: over 300 s of the full nonlinear simulator a 1.000 mm lateral offset returns 0.997 mm and a 0.500° yaw offset creeps to 0.565° at $+4.8 \times 10^{-5} \text{ }^\circ/\text{s}$, needing 3.3 h to reach the deadband edge.

Cross-validation, and scope. Random plant perturbations spanning two decades ($10^{-4} - 10^{-2}$) decay in the full nonlinear simulator (adaptive bias live, nothing frozen) at $\rho = 0.9951 - 0.9955$ against the 0.99530 predicted for the anchor-integral mode ($\tau = 2.12 \text{ s}$, 0.033 Hz) a generic perturbation excites: agreement to 2×10^{-4} . The certificate is local and on the nominal model, at zero delay and clean sensing; it estimates no region of attraction, and the push results of Section 5.2 lie outside it by construction, a push being what drives the gates off their flat regions. It establishes that the point ACC parks at is exponentially attracting in 55 of 57 directions and neutral in the other two, both named, mechanistically explained and separately bounded.

Table 5. Local stability certificate at the standing fixed point, across the five calibrated height variants. Central-difference linearization of the 100 Hz closed-loop step after a 60 s settle; “Flat” counts exported gates bit-exactly saturated over the last 20 s. ρ_{red} excludes the two centre directions, bounded separately by nonlinear probes.

Variant	CoM z (m)	Flat	Ctr.	Unst.	ρ_{red}	τ_{slow} (s)
Nominal	0.4069	19/21	2	0	0.99818	5.49
High small	0.4164	18/21	2	0	0.99787	4.70
High tiny	0.4128	19/21	2	0	0.99817	5.46
Low small	0.3973	19/21	2	0	0.99813	5.34
Low tiny	0.4009	19/21	2	0	0.99815	5.41

Produced by `scripts/acc_stability_certificate.py`; full spectra, gate traces and probe trajectories in the released output.

6. Discussion and Conclusion

Every gate in ACC was added after tracing a measured failure mode to its root cause: a bistable symmetric envelope, global integral windup, parametric pumping of a scheduled stiffness, hard-leash oscillation and self-referential stiffness. All thresholds were frozen before the ablation sweep, so Table 2 scores a fixed specification rather than a tuned one, and because the gating variables are physical quantities it runs without training, with a reduced sim-to-real gap hypothesised pending hardware validation (Grimminger et al., 2020).

Statistical power. Welch’s t -test at Bonferroni-adjusted $\alpha = 0.0056$ bounds the four subtractive removals rather than merely failing to reject them ($|d| \leq 0.49$, $< 1.4 \text{ N}$; Table 2); $N = 10$ success rates carry a Clopper–Pearson 95% CI of $[0.69, 1.00]$, the $N = 30$ push ablations resolve $\sim 1 \text{ N}$, and no claim rests on less.

Limitations. (1) *Simulation only.* Hardware validation remains future work. (2) *Deployment timing is the binding constraint.* ACC as tuned requires a 100 Hz loop and a latency clearing the 6–8 ms push-margin cliff of §5.6. Rate independence is not claimed: at 50 Hz with recomputed coefficients the platform does not stand, and the cause, whether plant-mode excitation or phase lag at $\Delta t = 20 \text{ ms}$, is unresolved. (3) *The calibrated height envelope is non-uniform.* A +10 cm height command during a push costs 69% of F_{max} , descent being free at every amplitude. (4) *The learning comparison is preliminary, and one classical family remains untested.* The model-free PPO run (5.4M steps, one seed, no domain randomisation (Tobin et al., 2017) or hyperparameter search) is a difficulty probe bounding nothing about RL here; the comparison this controller is built for, residual learning over the prior, is not run. Predictive control is the remaining untested classical family, but a receding horizon repairs only part of what fails here: with terminal cost set to that design’s Riccati solution, a convex MPC over the linearisation of §5.4.1 emits exactly its LQR input wherever no constraint is active (Rawlings et al., 2017), and through the entire pre-fall drift phase none is (2.0–3.2 Nm torque RMS against a 30 Nm bound). That drift is a static offset which state feedback without integral action cannot reject, and offset-free MPC rejects it by augmenting a disturbance state, i.e. by adding the integral action the anchor already is. Preview would buy something in the push regime, where the torque and $\pm 0.7 \text{ rad}$ hip-roll bounds do bind, and across the height band through relinearisation: the two comparisons left open. (5) *Generality is undemonstrated.* Every result is for one 10-DOF platform with one set of gains.

Deployment risks. ACC needs only raw torque, a 400 Nm/s rate limit, an IMU and encoders. The binding figure is the 6–8 ms latency cliff in push margin (§5.6); the stability threshold is twice as far out (station held through 14 ms), the ordering the architecture predicts. An estimated non-RTOS budget totals $\approx 8 \text{ ms}$: sensor readout $\sim 1 \text{ ms}$, state estimation $\sim 2 \text{ ms}$, actuation $\sim 3.5 \text{ ms}$, all assumed, plus a *measured* $\leq 1.5 \text{ ms}$

control step (p99 over five repeats of 5000 steps). The quantity that extrapolates to an embedded target is not host wall-clock but the operation count: 27 664 floating-point operations and 159 transcendentals per step, identical on every repeat, with no iterative solve, no matrix factorization and no data-dependent control flow (every gate of §4.3 is a branchless select), so worst-case execution time equals average-case, which a QP-based controller cannot offer. A 480 MHz Cortex-M7 retires that in ~ 0.07 ms; charged at 0.1 ms the estimate is 6.6 ms, so readout and actuation decide the matter, not compute. Two mitigations compose: a real-time OS recovers 2–4 ms, and doubling the scheduled sagittal damping $s_v k_{vel}$ moves the knee to 8–10 ms. Hardware repeatability is projected at 1–3 mm rather than 0.294 mm.²

Conclusion. ACC is a two-channel torque assembly in which a PD balance core is augmented by a proximity-gated anchor integral. It reaches 0.294 ± 0.015 mm idle sagittal repeatability, a $59\times$ improvement over the proportional-only baseline ($N=10$, Table 1), with omnidirectional push survival from $F_{min}=82$ N to a 108 N median; absolute accuracy is separate: the robot parks 27.0 mm from commanded home and holds that stand-off to 0.009 mm, matching a closed-form model to 0.45 %. Six classical baselines and a full-state LQR derived from the plant itself all fall, and integral action does not prevent it. The contribution is a *structured classical prior*: an interpretable controller whose gating encodes the plant’s measured failure modes. The residual stack it serves is future work, with hardware validation, a resourced learning comparison, and conditional activation for locomotion.

Code and Data Availability

Controller, MuJoCo model, evaluation scripts and raw results are at <https://github.com/thuong180702/Wheeled-bipedal-robot-simulation>; each table’s note names its script, and the extended version there carries the full derivations, WBC details and parking-offset model.

31 supplementary clips are under `paper/videos/`, rendered by `scripts/render_paper_videos.py` from the same harness and seeds as the corresponding table: quiet stance (1); push recovery (10, including a deliberate 85 N oblique *fall* bracketing the weakest bearing of Table 2 from above); drops 10–100 cm (6); one-wheel

curbs (3); ramp and ledge 20–50 cm (8); height transitions (3). Each is one trial at one seed, illustrating a regime rather than establishing a rate; that directory’s `README.md` gives the per-clip protocol.

7. Notes on contributors



Author’s name is an independent researcher based in Viet Nam, working on classical and learning-based control for wheeled bipedal platforms, whole-body control, and structured model-based priors for residual rein-

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²Produced by `scripts/measure_control_step_cost.py`.

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